PROJECT REPORT

AWS VPC PEERING

Communication Between Two Amazon VPCs

Submitted to: Team EWB Courses

Project: AWS Cloud Computing Internship

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1. Project Overview

This project demonstrates how to connect two Amazon Virtual Private Clouds (VPCs) using VPC Peering in Amazon Web Services (AWS). The connection allows resources in the two VPCs to communicate using private IP addresses when the required routes and security rules are configured.

2. Project Objectives

- Create two VPCs with different, non-overlapping IP address ranges.
- Create public and private subnets.
- Create and activate a VPC Peering connection between the VPCs.
- Configure route tables so traffic can travel through the peering connection.
- Configure security groups to allow only the required traffic.
- Launch EC2 instances and test private connectivity in both directions.
- Test how an ICMP security rule affects ping connectivity.

3. Introduction

Amazon VPC is a logically isolated virtual network in AWS where cloud resources can be launched. A subnet is a smaller IP network inside a VPC. A route table controls where network traffic is sent. A security group acts as a virtual firewall for an instance.

VPC Peering is a network connection between two VPCs. In this project, VPC A and VPC B were connected so that the application instances could communicate using private IP addresses without requiring public IP addresses on those application instances.

4. AWS Services and Tools Used

- Amazon VPC — to create isolated virtual networks.
- VPC Peering — to connect VPC A and VPC B.
- Amazon EC2 — to run the Bastion, App-A, and App-B instances.
- Route Tables — to direct traffic to the local network, Internet Gateway, or peering connection.
- Internet Gateway — to provide internet routing for the public subnet in VPC A.
- Security Groups — to control SSH and ICMP traffic.
- Amazon Linux 2023 and the ping command — for instance access and connectivity testing.

5. Network Design and IP Addressing

5.1 VPC and Subnet Details

Resource	Name	CIDR block
VPC A	ewb-vpc-a	10.10.0.0/16
VPC A public subnet	ewb-a-public	10.10.1.0/24
VPC A private subnet	ewb-a-private	10.10.2.0/24
VPC B	ewb-vpc-b	10.20.0.0/16

VPC B private subnet	ewb-b-private	10.20.1.0/24

5.2 EC2 Instance Details

Instance	Private IP address	Purpose
ewb-bastion	10.10.1.10	Entry point to reach private instances
ewb-app-a	10.10.2.10	Application instance in VPC A
ewb-app-b	10.20.1.10	Application instance in VPC B

The application instances were configured without public IPv4 addresses. Access to the private instances was performed through the Bastion host.

6. Implementation Steps

1. Created VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16) in the Mumbai AWS Region.
2. Created the public subnet and private subnet in VPC A, and a private subnet in VPC B.
3. Created and attached an Internet Gateway to VPC A for the public subnet.
4. Created the required route tables and associated them with their subnets.
5. Created a VPC Peering connection between VPC A and VPC B and accepted the request. The connection reached the Active state.
6. Added a route to 10.20.0.0/16 through the peering connection in the VPC A private route table.
7. Added a route to 10.10.0.0/16 through the peering connection in the VPC B private route table.
8. Configured security groups to permit required SSH access and ICMP testing traffic.
9. Launched the Bastion, App-A, and App-B EC2 instances using Amazon Linux 2023.
10. Connected to the private instances through the Bastion host and performed ping tests.

7. Route Table Configuration

7.1 VPC A Private Route Table

The route table contains the local route for 10.10.0.0/16 and a route for 10.20.0.0/16 pointing to the VPC Peering connection.

7.2 VPC B Private Route Table

The route table contains the local route for 10.20.0.0/16 and a route for 10.10.0.0/16 pointing to the VPC Peering connection.

7.3 VPC A Public Route Table

The public route table contains a route to the Internet Gateway for internet-bound traffic and a route for 10.20.0.0/16 through the VPC Peering connection.

8. Security Configuration

- The Bastion security group allowed SSH (TCP port 22) from the user's permitted public IP address.

- The App-A and App-B security groups allowed SSH from the Bastion host's private IP address, as configured for the lab.
- ICMP traffic was permitted between App-A and App-B for ping testing.
- Default outbound rules were retained for the lab.

Security groups are stateful virtual firewalls. Access should be limited to required source addresses and ports. The SSH source rules should not be opened to all IP addresses unless there is a specific, temporary lab need.

9. Testing and Results

9.1 Test from App-A to App-B

Command used: ping -c 4 10.20.1.10

Result: Successful. Four packets were transmitted and four were received, with 0% packet loss.

9.2 Test from App-B to App-A

Command used: ping -c 4 10.10.2.10

Result: Successful. Four packets were transmitted and four were received, with 0% packet loss.

9.3 Controlled ICMP Failure Test

The inbound ICMP rule on the App-B security group was temporarily removed. The ping test from App-A then failed with 100% packet loss. This showed that the security group rule was required for ping traffic to reach App-B.

9.4 Connectivity Restored

The ICMP rule was restored and the ping test was run again. Connectivity succeeded with 0% packet loss.

10. Screenshots and Evidence

The GitHub repository contains screenshots showing the VPCs and CIDR blocks, subnets, route tables, active VPC Peering connection, security groups, EC2 instances, SSH access, successful ping tests in both directions, the blocked ICMP test, and the restored connectivity test.

GitHub repository: https://github.com/Abhishek-7607/AWS-VPC-Peering-Project

Screenshots folder: screenshots/

Before submission, confirm that all screenshots are present in the screenshots folder and that the repository link is accessible to the trainer.

11. Challenges and Learning

- Learned how VPCs and subnets use CIDR blocks to separate network ranges.

- Learned that VPC Peering requires routes in both VPCs for two-way communication.
- Learned that security group rules can allow or block network traffic even when routes are configured.
- Practiced accessing private EC2 instances through a Bastion host.
- Used ping tests to verify connectivity and troubleshoot security-rule issues.

12. Conclusion

The AWS VPC Peering project was successfully configured and tested. VPC A and VPC B communicated using the private IP addresses of App-A and App-B. Ping tests succeeded in both directions. Temporarily removing the required ICMP rule blocked ping traffic, and restoring the rule restored connectivity. The project provided practical experience with AWS VPC networking, routing, security groups, EC2 instances, and basic network troubleshooting.

13. Submission Details

Project: AWS VPC Peering

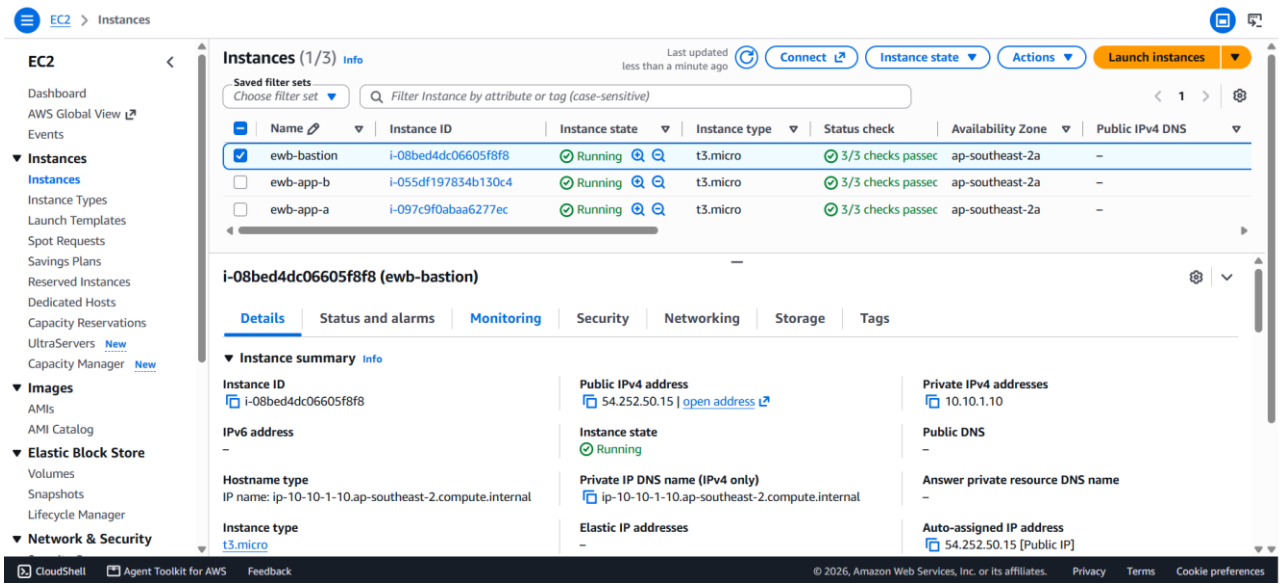
Program: AWS Cloud Computing Internship

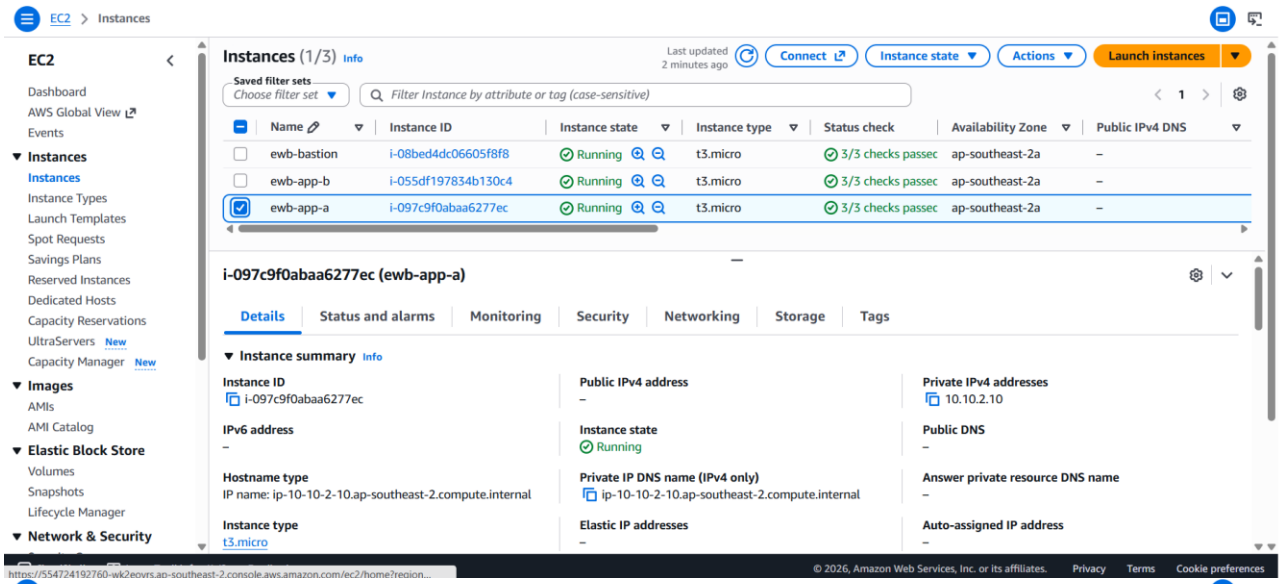
Organization: Team EWB Courses

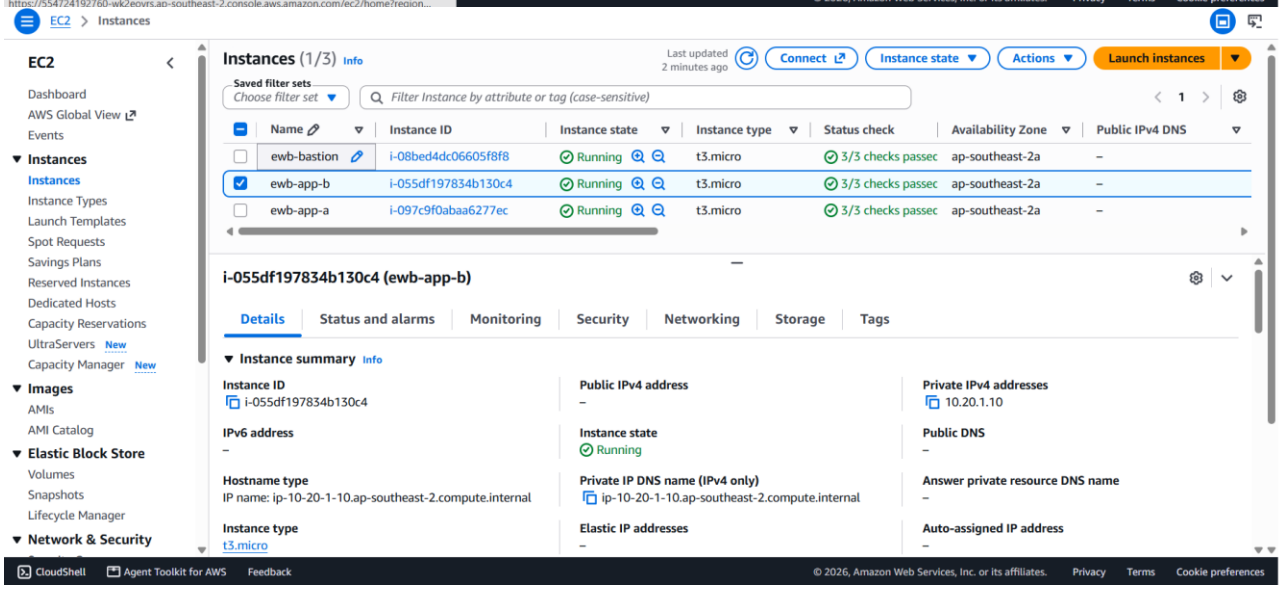
Student Name: Abhishek Kumar Kushwaha

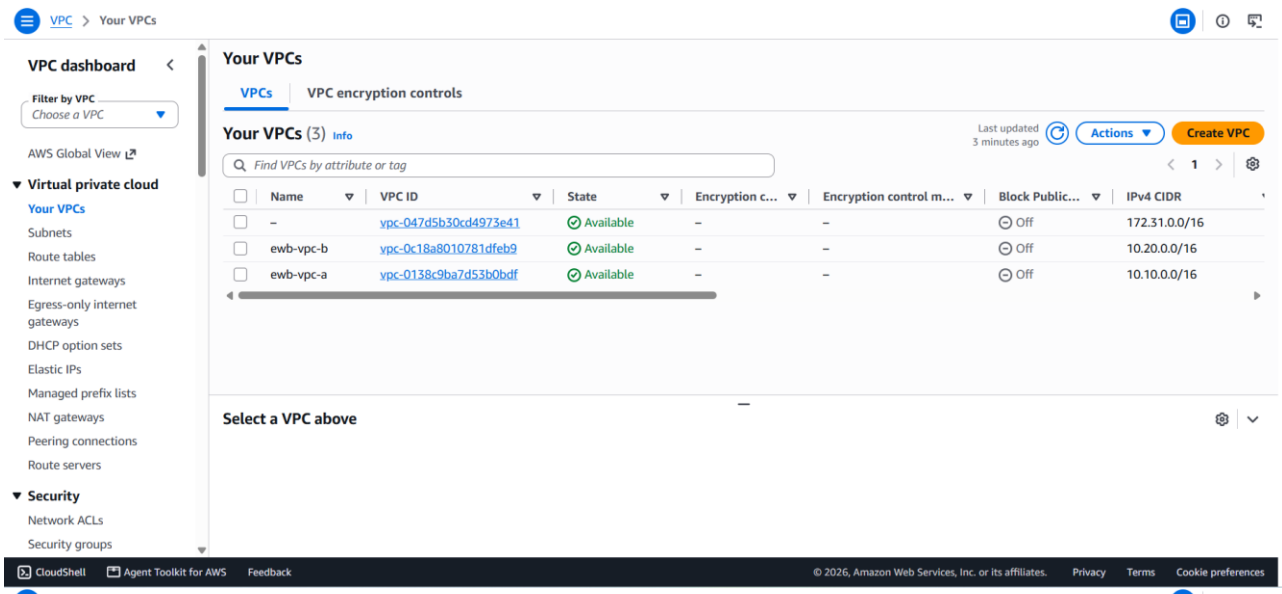
AICTE Student ID: STU5577859773cc51789473748

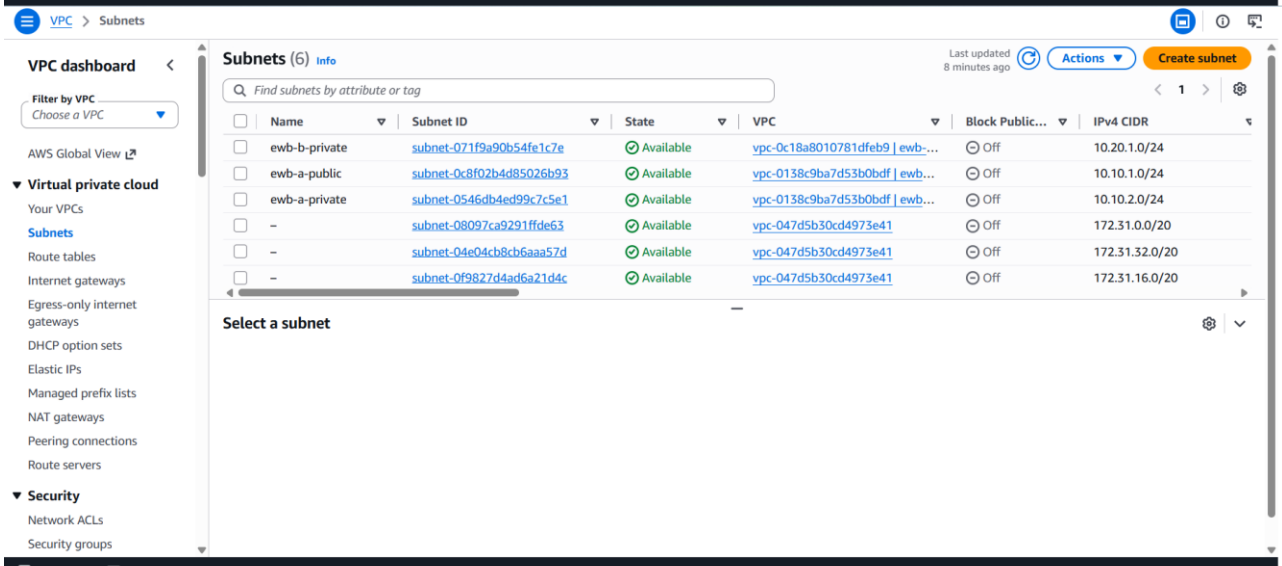
Date:19/10/2026

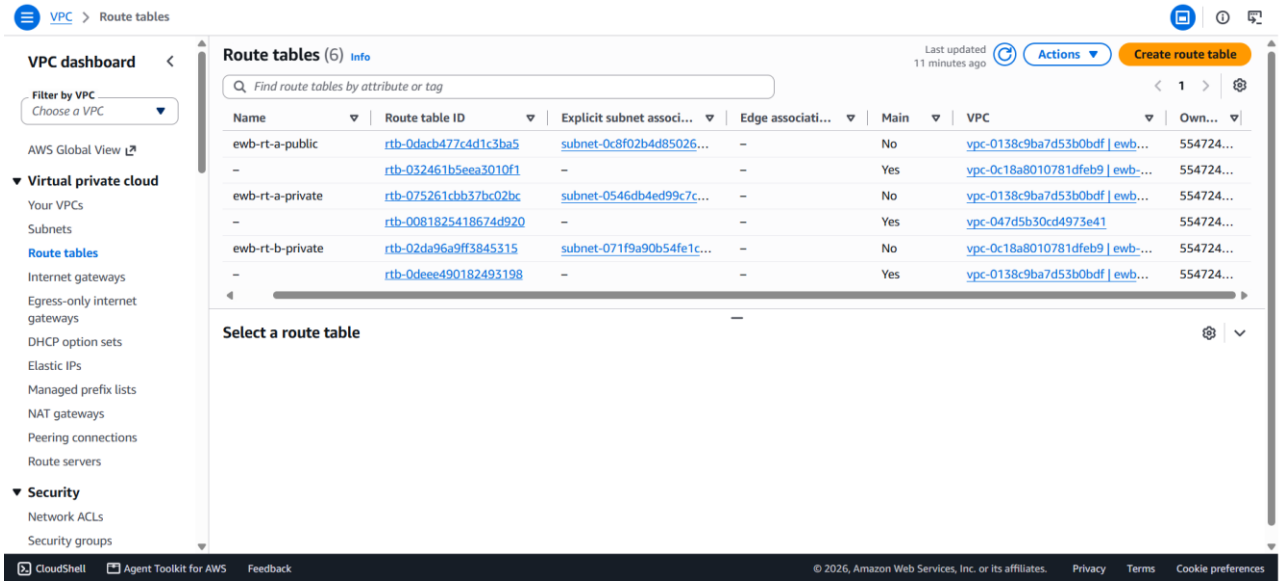

EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Instances (1/3) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
ewb-bastion i-08bed4dc06605f8f8 Running t3.micro 3/3 checks passec ap-southeast-2a –
ewb-app-b i-055df197834b130c4 Running t3.micro 3/3 checks passec ap-southeast-2a –
ewb-app-a i-097c9f0abaa6277ec Running t3.micro 3/3 checks passec ap-southeast-2a –
i-08bed4dc06605f8f8 (ewb-bastion)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-08bed4dc06605f8f8
Public IPv4 address
54.252.50.15 | open address
Private IPv4 addresses
10.10.1.10
IPv6 address
–
Instance state
Running
Public DNS
–
Hostname type
IP name: ip-10-10-1-10.ap-southeast-2.compute.internal
Private IP DNS name (IPv4 only)
ip-10-10-1-10.ap-southeast-2.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
54.252.50.15 [Public IP]
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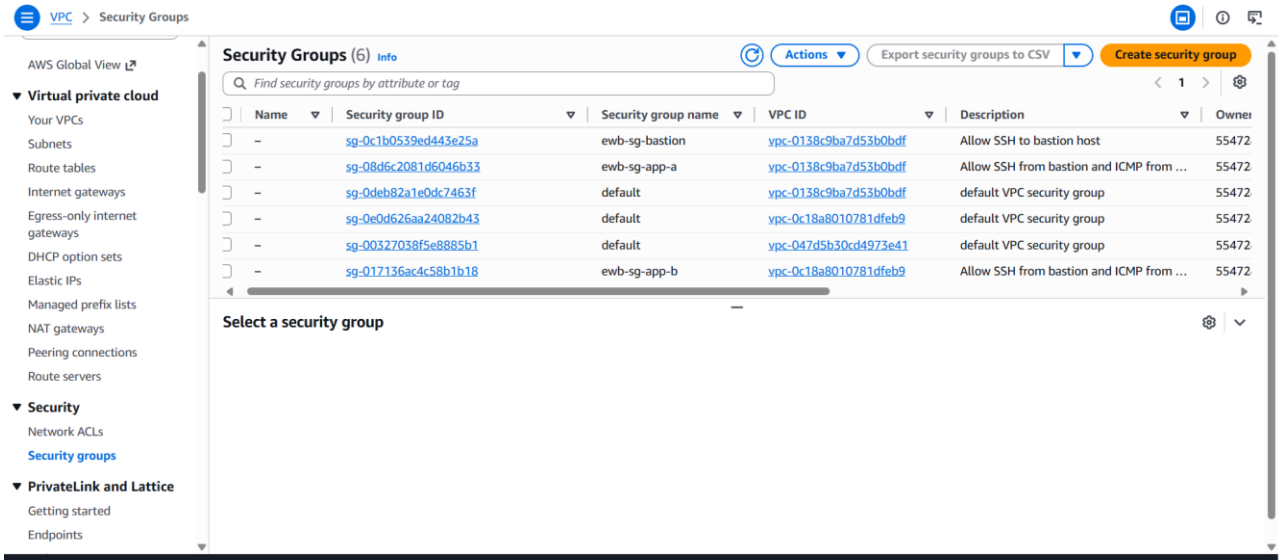

EC2 > Instances
Instances (1/3) Info
Last updated 2 minutes ago
Connect
Instance state
Actions
Launch instances
ewb-bastion i-08bed4dc06605f8f8 Running t3.micro 3/3 checks passec ap-southeast-2a –
ewb-app-b i-055df197834b130c4 Running t3.micro 3/3 checks passec ap-southeast-2a –
ewb-app-a i-097c9f0abaa6277ec Running t3.micro 3/3 checks passec ap-southeast-2a –
i-097c9f0abaa6277ec (ewb-app-a)
Details
Status and alarms
Monitoring
Security
Networking
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Tags
Instance summary Info
Instance ID
i-097c9f0abaa6277ec
Public IPv4 address
–
Private IPv4 addresses
10.10.2.10
IPv6 address
–
Instance state
Running
Public DNS
–
Hostname type
IP name: ip-10-10-2-10.ap-southeast-2.compute.internal
Private IP DNS name (IPv4 only)
ip-10-10-2-10.ap-southeast-2.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
–
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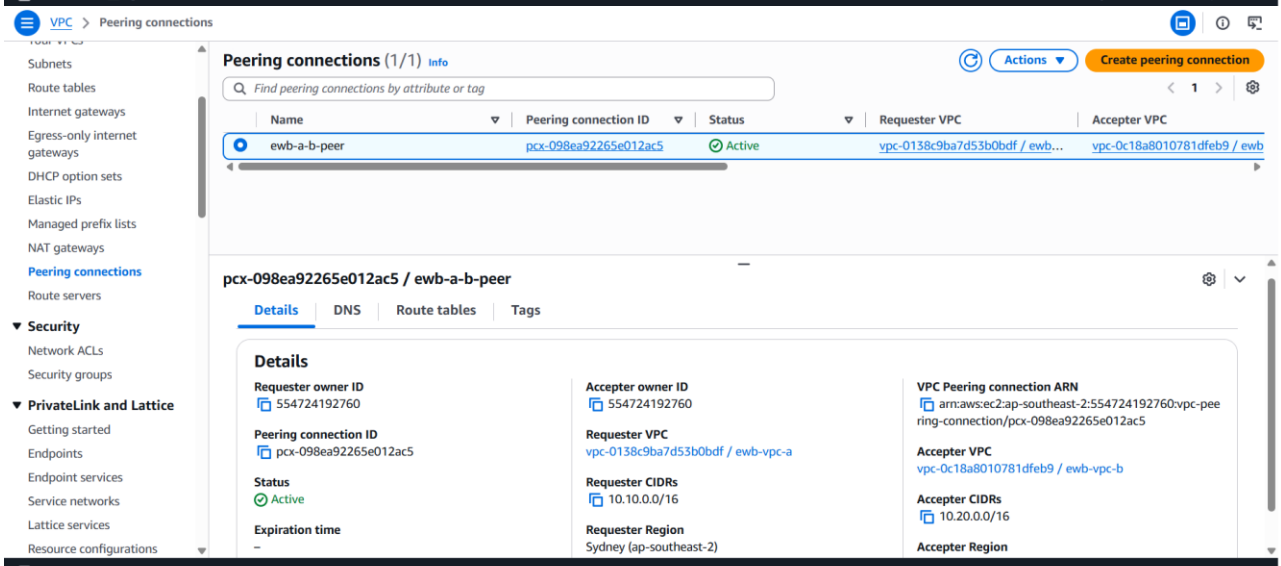

EC2 > Instances
Instances (1/3) Info
Last updated 2 minutes ago
Connect
Instance state
Actions
Launch instances
ewb-bastion i-08bed4dc06605f8f8 Running t3.micro 3/3 checks passec ap-southeast-2a –
ewb-app-b i-055df197834b130c4 Running t3.micro 3/3 checks passec ap-southeast-2a –
ewb-app-a i-097c9f0abaa6277ec Running t3.micro 3/3 checks passec ap-southeast-2a –
i-055df197834b130c4 (ewb-app-b)
Details
Status and alarms
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Tags
Instance summary Info
Instance ID
i-055df197834b130c4
Public IPv4 address
–
Private IPv4 addresses
10.20.1.10
IPv6 address
–
Instance state
Running
Public DNS
–
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IP name: ip-10-20-1-10.ap-southeast-2.compute.internal
Private IP DNS name (IPv4 only)
ip-10-20-1-10.ap-southeast-2.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
–
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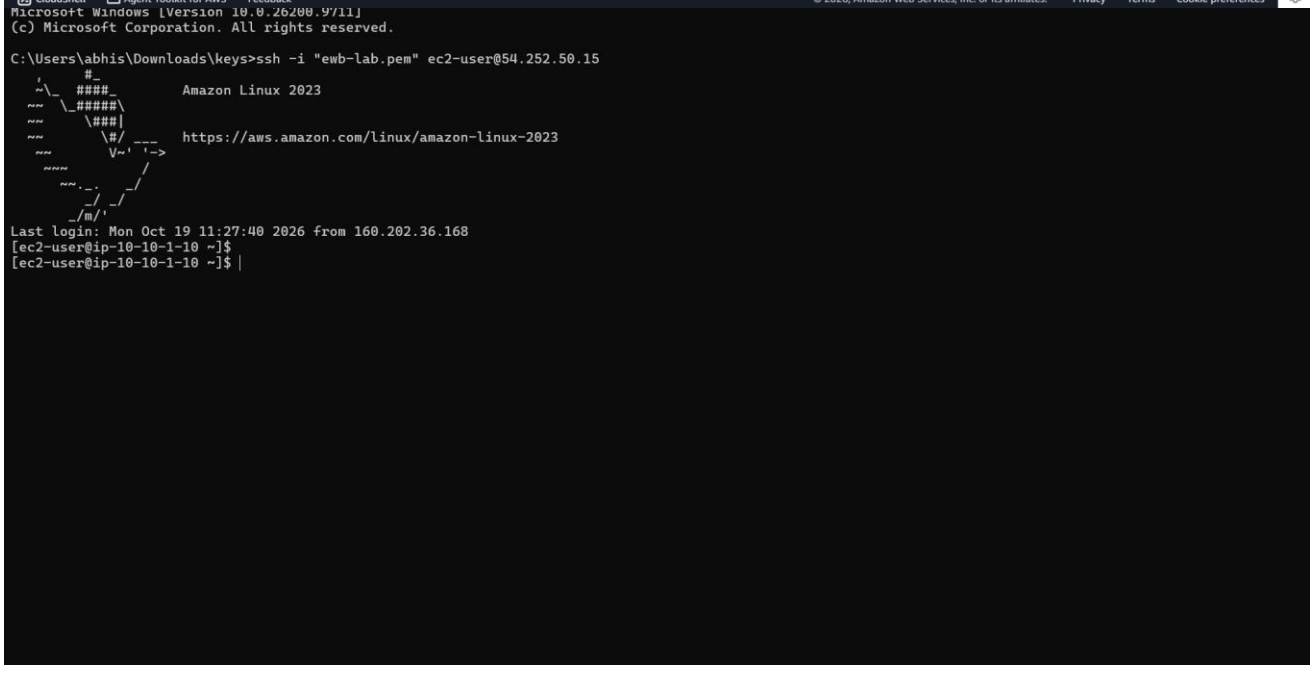

VPC > Your VPCs
VPC dashboard
Filter by VPC
Choose a VPC
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Virtual private cloud
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Egress-only internet gateways
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NAT gateways
Peering connections
Route servers
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Your VPCs
VPCs
VPC encryption controls
Your VPCs (3) Info
Last updated 3 minutes ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control m...
Block Public...
IPv4 CIDR
– vpc-047d5b30cd4973e41 Available – – Off 172.31.0.0/16
ewb-vpc-b vpc-0c18a8010781dfeb9 Available – – Off 10.20.0.0/16
ewb-vpc-a vpc-0138c9ba7d53b0bdf Available – – Off 10.10.0.0/16
Select a VPC above
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VPC > Subnets
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Security groups
Subnets (6) Info
Last updated 8 minutes ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
Block Public...
IPv4 CIDR
ewb-b-private subnet-071f9a90b54fe1c7e Available vpc-0c18a8010781dfeb9 | ewb-... Off 10.20.1.0/24
ewb-a-public subnet-0c8f02b4d85026b93 Available vpc-0138c9ba7d53b0bdf | ewb... Off 10.10.1.0/24
ewb-a-private subnet-0546db4ed99c7c5e1 Available vpc-0138c9ba7d53b0bdf | ewb... Off 10.10.2.0/24
– subnet-08097ca9291ffde63 Available vpc-047d5b30cd4973e41 Off 172.31.0.0/20
– subnet-04e04cb8cb6aaa57d Available vpc-047d5b30cd4973e41 Off 172.31.32.0/20
– subnet-0f9827d4ad6a21d4c Available vpc-047d5b30cd4973e41 Off 172.31.16.0/20
Select a subnet
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Route tables (6) Info
Last updated 11 minutes ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associati...
Main
VPC
Own...
ewb-rt-a-public rtb-0dacb477c4d1c3ba5 subnet-0c8f02b4d85026... – No vpc-0138c9ba7d53b0bdf | ewb... 554724...
– rtb-032461b5eea3010f1 – – Yes vpc-0c18a8010781dfeb9 | ewb-... 554724...
ewb-rt-a-private rtb-075261cbb37bc02bc subnet-0546db4ed99c7c... – No vpc-0138c9ba7d53b0bdf | ewb... 554724...
– rtb-0081825418674d920 – – Yes vpc-047d5b30cd4973e41 554724...
ewb-rt-b-private rtb-02da96a9ff3845315 subnet-071f9a90b54fe1c... – No vpc-0c18a8010781dfeb9 | ewb-... 554724...
– rtb-0deee490182493198 – – Yes vpc-0138c9ba7d53b0bdf | ewb... 554724...
Select a route table
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Endpoints
Security Groups (6) Info
Actions
Export security groups to CSV
Create security group
Find security groups by attribute or tag
Name
Security group ID
Security group name
VPC ID
Description
Owner
– sg-0c1b0539ed443e25a ewb-sg-bastion vpc-0138c9ba7d53b0bdf Allow SSH to bastion host 55472
– sg-08d6c2081d6046b33 ewb-sg-app-a vpc-0138c9ba7d53b0bdf Allow SSH from bastion and ICMP from ... 55472
– sg-0deb82a1e0dc7463f default vpc-0138c9ba7d53b0bdf default VPC security group 55472
– sg-0e0d626aa24082b43 default vpc-0c18a8010781dfeb9 default VPC security group 55472
– sg-00327038f5e8885b1 default vpc-047d5b30cd4973e41 default VPC security group 55472
– sg-017136ac4c58b1b18 ewb-sg-app-b vpc-0c18a8010781dfeb9 Allow SSH from bastion and ICMP from ... 55472
Select a security group
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Lattice services
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Peering connections (1/1) Info
Actions
Create peering connection
Find peering connections by attribute or tag
Name
Peering connection ID
Status
Requester VPC
Accepter VPC
ewb-a-b-peer pcx-098ea92265e012ac5 Active vpc-0138c9ba7d53b0bdf / ewb... vpc-0c18a8010781dfeb9 / ewb
pcx-098ea92265e012ac5 / ewb-a-b-peer
Details
DNS
Route tables
Tags
Details
Requester owner ID
554724192760
Peering connection ID
pcx-098ea92265e012ac5
Status
Active
Expiration time
–
Accepter owner ID
554724192760
Requester VPC
vpc-0138c9ba7d53b0bdf / ewb-vpc-a
Requester CIDRs
10.10.0.0/16
Requester Region
Sydney (ap-southeast-2)
VPC Peering connection ARN
arn:aws:ec2:ap-southeast-2:554724192760:vpc-peering-connection/pcx-098ea92265e012ac5
Accepter VPC
vpc-0c18a8010781dfeb9 / ewb-vpc-b
Accepter CIDRs
10.20.0.0/16
Accepter Region
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Microsoft Windows [Version 10.0.26200.9711]
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C:\Users\abhis\Downloads\keys>ssh -i "ewb-lab.pem" ec2-user@54.252.50.15
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Mon Oct 19 11:27:40 2026 from 160.202.36.168
[ec2-user@ip-10-10-1-10 ~]$
[ec2-user@ip-10-10-1-10 ~]$

```
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 19 11:19:48 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping 10.20.1.10
```

```
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 19 11:19:06 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$
```

```
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 19 11:19:48 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.272 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.262 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.264 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.252 ms
64 bytes from 10.20.1.10: icmp_seq=5 ttl=127 time=0.272 ms
64 bytes from 10.20.1.10: icmp_seq=6 ttl=127 time=0.249 ms
64 bytes from 10.20.1.10: icmp_seq=7 ttl=127 time=0.250 ms
64 bytes from 10.20.1.10: icmp_seq=8 ttl=127 time=0.261 ms
64 bytes from 10.20.1.10: icmp_seq=9 ttl=127 time=0.259 ms
64 bytes from 10.20.1.10: icmp_seq=10 ttl=127 time=0.248 ms
64 bytes from 10.20.1.10: icmp_seq=11 ttl=127 time=0.282 ms
^C
--- 10.20.1.10 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10426ms
rtt min/avg/max/mdev = 0.248/0.261/0.282/0.010 ms
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
```

```
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 19 11:19:06 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.246 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.261 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.252 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.274 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.246/0.258/0.274/0.010 ms
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
[ec2-user@ip-10-20-1-10 ~]$
```

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3138ms

[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
[ec2-user@ip-10-10-2-10 ~]$
```